ADVANCED IP SUBNETTING

9th March 2026

NIAI 2nd Assignment

Scenario 1 – The Small Law Firm

> *Network: 192.168.15.0/24. Need exactly 4 subnets: Partners, Associates, Paralegals, Guest Wi-Fi.*

A) New subnet mask (CIDR and decimal)

Step 1 – Determine borrowed bits: 4 subnets require $2^n \geq 4$. $2^2 = 4 \rightarrow$ borrow 2 bits.

Step 2 – New CIDR: /24 + 2 = `/26`.

Step 3 – Decimal mask: First three octets 255.255.255. Fourth octet: first 2 bits set to 1 → binary 11000000 = 192. So mask = `255.255.255.192`.

> Block size (increment) = 256 − 192 = 64. Subnets: .0, .64, .128, .192.

B) Usable hosts per subnet

Host bits: 32 − 26 = 6 bits.

> Usable hosts = $2^6 - 2 = 64 - 2 =$ **62**.

C) 2nd subnet (Associates) – Network ID, usable range, broadcast

Subnets (0-based): 0: .0, 1: .64, 2: .128, 3: .192.

2nd subnet = index 1: Network ID = `192.168.15.64`.

Broadcast: Next subnet network = .128 → broadcast = .128 − 1 = `192.168.15.127`.

Usable range: first address after network (.65) to last before broadcast (.126).

Network ID	192.168.15.64
Usable IP range	192.168.15.65 – 192.168.15.126
Broadcast address	192.168.15.127

Scenario 2 – The Medical Clinic

> *Network: 200.50.25.0/24. Need minimum 6 subnets.*

A) Bits to borrow and total subnets

Find smallest n where $2^n \geq 6$: $2^2=4$ (too few), $2^3=8$ → borrow 3 bits.

> Total subnets created = $2^3 = 8$.

B) New subnet mask

CIDR: /24 + 3 = `/27`.

Decimal: Fourth octet: first 3 bits = 1 → 11100000 = 224 → mask `255.255.255.224`.

> Block size = 256 − 224 = 32.

C) 1st subnet (Reception) – Network ID and broadcast

Subnet 0: Network ID = `200.50.25.0`.

Broadcast: Next subnet network = .32 → broadcast = .31 → `200.50.25.31`.

Network ID	200.50.25.0
Broadcast	200.50.25.31

Scenario 3 – High School Computer Labs

> *Network: 192.168.100.0/24. 4 subnets, each must support at least 50 computers.*

A) Subnet mask allowing 4 subnets with ≥50 hosts

Host requirement: $2^h - 2 \geq 50$. $2^5=32$ → usable 30 (too low). $2^6=64$ → usable 62 (sufficient). So need at least 6 host bits.

Subnet requirement: 4 subnets → $2^s \geq 4$ → s = 2 (since $2^2=4$).

Check total: s + h = 2 + 6 = 8, which fits in a /24.

New CIDR: /24 + 2 = `/26` → decimal `255.255.255.192`.

B) Maximum usable hosts per subnet

> $2^6 - 2 = 64 - 2 =$ **62**.

C) Usable IP range for the 4th subnet

Subnets (block 64): .0, .64, .128, .192. 4th subnet = index 3 → network `192.168.100.192`.

Broadcast: Next subnet would be .256, so broadcast = .255.

Usable range: .193 to .254.

Usable range	192.168.100.193 - 192.168.100.254

Scenario 4 – The Coffee Shop Chain

> *Network: 198.100.50.0/24. Maximum number of subnets, each with at least 14 usable hosts.*

A) Most efficient subnet mask

Find minimum host bits for ≥14 hosts: $2^3=8$ → usable 6 (too few). $2^4=16$ → usable 14 (exact). So h = 4 host bits.

Subnet bits: borrowed = 8 − 4 = 4 → CIDR = /24 + 4 = `/28`.

Decimal mask: fourth octet: 11110000 = 240 → `255.255.255.240`.

> Block size = 256 − 240 = 16.

B) Total subnets created

> 2^4 = **16** subnets.

C) Network ID for the 3rd subnet

Subnet numbering (0-based, increment 16): 0→.0, 1→.16, 2→.32, 3→.48 …

3rd subnet = index 2 → network `198.100.50.32`.

Scenario 5 – The E-Commerce Startup

> *Network: 203.0.113.0/24. Need 12 distinct subnets.*

A) Bits to borrow

> $2^3=8$ (<12), $2^4=16$ (≥12) → borrow `4 bits`.

B) Usable IP addresses per subnet

Host bits = 8 − 4 = 4 → usable = $2^4 - 2 = 16 - 2 =$ **14**.

C) Last (highest) subnet details

With 4 borrowed bits, total subnets = 16 (indices 0–15). Last subnet index = 15.

Network = 203.0.113.0 + 15×16 = 203.0.113.240.

Broadcast = next subnet would be .256 → broadcast = .255.

Usable range = .241 to .254.

Network ID	`203.0.113.240`
Usable IP range	`203.0.113.241 - 203.0.113.254`
Broadcast address	`203.0.113.255`

Scenario 6 – The Regional Hospital

> *Network: 172.16.0.0/16. Need 50 separate subnets.*

A) Bits borrowed and new mask

> $2^5=32$ (<50), $2^6=64$ → borrow 6 bits.

CIDR = /16 + 6 = `/22`.

Decimal: third octet: first 6 bits = 1 → 11111100 = 252 → mask `255.255.252.0`.

> Block size in 3rd octet = 256 − 252 = 4.

B) Usable hosts per subnet

Host bits = 32 − 22 = 10 → usable = $2^{10} - 2 = 1024 - 2 =$ **1022**.

C) 2nd subnet – Network ID and broadcast

Subnets: 172.16.0.0/22, 172.16.4.0/22, 172.16.8.0/22, …

2nd subnet = index 1 → network `172.16.4.0`.

Broadcast = next subnet (172.16.8.0) minus 1 → `172.16.7.255`.

Network ID	172.16.4.0
Broadcast	172.16.7.255

Scenario 7 – The Corporate Headquarters

> *Network: 172.25.0.0/16. Each subnet must support exactly 500 usable hosts.*

A) Host bits required

> $2^8-2 = 254$ (<500), $2^9-2 = 510$ (≥500) → host bits = 9.

B) New subnet mask

Borrowed bits = 16 − 9 = 7 → CIDR = /16 + 7 = `/23`.

Decimal: third octet: first 7 bits = 1 → 11111110 = 254 → mask `255.255.254.0`.

C) Block size (magic number) in the 3rd octet

> 256 − 254 = `2`.

Scenario 8 – The Smart City Project

> *Network: 128.10.0.0/16. Need 100 subnets.*

A) Required subnet mask

> $2^6=64$ (<100), $2^7=128$ → borrow 7 bits → CIDR = /23 → mask `255.255.254.0`.

B) Usable IP addresses per subnet

Host bits = 32 − 23 = 9 → usable = $2^9 - 2 = 512 - 2 =$ **510**.

C) Network ID for the 3rd subnet

Block size = 2 in third octet. Subnets: 128.10.0.0, 128.10.2.0, 128.10.4.0, …

3rd subnet = index 2 → network `128.10.4.0`.

Scenario 9 – The University Wi-Fi Zones

> *Network: 172.31.0.0/16. Each subnet must support at least 2,000 hosts.*

A) Required subnet mask

> $2^{10}-2 = 1022$ (<2000), $2^{11}-2 = 2046$ (≥2000) → host bits = 11 → borrowed = 5 → CIDR = /21.

Decimal: third octet: first 5 bits = 1 → 11111000 = 248 → mask `255.255.248.0`.

B) Total subnets allowed

> $2^5 =$ **32** subnets.

C) Usable IP range for the 1st subnet

Block size = 256 − 248 = 8 in third octet. 1st subnet: 172.31.0.0/21.

Broadcast = next subnet (172.31.8.0) minus 1 = 172.31.7.255.

Usable range: 172.31.0.1 to 172.31.7.254.

Usable range	`172.31.0.1 - 172.31.7.254`

Scenario 10 – The Manufacturing Plant

> *Network: 150.20.0.0/16. Need exactly 16 equal subnets.*

A) Subnet mask

> $2^4 = 16$ → borrow 4 bits → CIDR = /20 → third octet: 11110000 = 240 → mask `255.255.240.0`.

B) Usable hosts per assembly line

Host bits = 32 − 20 = 12 → usable = $2^{12} - 2 = 4096 - 2 =$ **4094**.

C) 5th subnet details

Block size = 16 in third octet. Subnets: .0, .16, .32, .48, .64, .80, …

5th subnet = index 4 → network `150.20.64.0`.

Broadcast = next subnet (150.20.80.0) minus 1 = `150.20.79.255`.

Usable range: 150.20.64.1 to 150.20.79.254.

Network ID	150.20.64.0
Usable IP range	150.20.64.1 – 150.20.79.254
Broadcast address	150.20.79.255